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Studierichting: Informatica

Oefeningenreeks 3A nr. 39.9

$$f(x) = (\sin x)^{\cos x}$$

$$\Rightarrow y = (\sin x)^{\cos x}$$

$$\Rightarrow \ln y = \ln ((\sin x)^{\cos x}) = (\cos x) \ln (\sin x)$$

$$\Rightarrow (\ln y)' = ((\cos x) \ln (\sin x))'$$

$$\Rightarrow \frac{y'}{y} = (-\sin x) \ln (\sin x) + (\cos x) \frac{\cos x}{\sin x}$$

$$\Rightarrow \frac{y'}{y} = -\sin x \ln (\sin x) + \cos x \frac{\cos x}{\sin x}$$

$$\Rightarrow y' = y \left(-\sin x \ln (\sin x) + \frac{\cos^2 x}{\sin x} \right)$$

$$\Rightarrow f'(x) = (\sin x)^{\cos x} \left(-\sin x \ln (\sin x) + \frac{\cos^2 x}{\sin x} \right)$$

References